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G. NARAYANAMMA INSTITUTE OF TECHNOLOGY & SCIENCE

AUTONOMOUS

(For Women)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

II B.Tech. I Semester (2024-25)

Database Management Systems Lab

(Course Code: 123A3, GNITS-R-22)

Date: 01/08/2024

LAB CYCLE – 1

Assignment-1

DDL and DML commands

1. Creating tables for various relations (in SQL).
2. Inserting data into relational tables.
3. Integrity Constraints (Table Level and Column Level).

SIMPLE QUERIES

Write and execute the following Queries on Employee and Department tables.

1. Display the details of employee names, salary, commission, hire date of employees.
2. Display the details of employees who are working in the department whose number is 20.
3. Display the details of employees whose salary is between 1000 and 3000.
4. Display the details of employees whose salary is less than 2000 or more than 3000.
5. Find all the employees who don't have a Manager.
6. Display the employee names and yearly salary with the header "Annual Salary".
7. Display the list of employees whose names starts with 'S' and contains exactly 5 characters.
8. Display the names of all employees, who have 'TH' or 'LL' in their names.
9. Display the list of employees who are working in dept 20 or getting salary more than 1000.
10. Find all clerks who earn between 1000 and 2000.
11. Find all the employees who have comm.
12. List department no and name in alphabetical order of department name.
13. List the details of the employees in department 10 and 20 in alphabetical order of name.
14. Display name and total remuneration of all employees.
15. Display all the employees who were hired during 1983.
16. Display name, annual salary and commission of all sales persons whose monthly salary is greater than their commission. The output should be in the descending order of salary. If two or more employees have the same salary sort by ename, with in the highest salary order.
17. Increase the salary for all employees by 15%.
18. Display the list of employees in the descending order of job and salaries

19. Display the Employee data as displayed below.

Who, what and when

Allen has held the position of salesman in dept 30 since 20-feb-81

Ward has held the position of salesman in dept 30 since 22-feb-81

20. Display the names of employees who are working as clerk, salesman or analyst and drawing a salary more than 2000.

21. **Display the list of employees who have joined the company before 30th June 90 or after 31st dec 90.**

Assignment-2 DRL Commands

1. Find out how many managers are there without listing them.
2. Find the difference of highest and lowest salaries for each job type.
3. Find the Minimum, Maximum and average salary for each job.
4. List the employee name, department name for each employee.
5. Display the list of employees who are working in department no 10 and whose salary is ≥ 5000 .
6. Display the list of employees whose name end with letter say 's'.
7. Display the list of employees whose salary is greater than 5000 and also have a commission.
8. Display the list of employees whose salary is less than 1000 and don't have any commission.
9. Display the list of employees who are not working in department 10.
10. Display the list of employees with their manager names.
11. **Display the total salary being paid to all employees.**
12. **Display name, Sal, hra, pf, da, total Sal for each employee. The output should be in the order of total Sal, hra 15% of Sal, da 10% of sal, pf 5% of sal total salary will be (sal*hra*da)-pf.**
13. **Display dept numbers and total number of employees within each group.**

Assignment-3 SQL Functions

1. Find the square root of 25.
2. Find the value of 2^3 .
3. Find the remainder of 11 divided by 3.
4. Find the value 3/11(round the result up to 2 digits after the decimal).
5. Round the value of 45.789 to the lowest integer, nearest largest integer.
6. Display the current date in the following format: 26th January 1998.
7. Find the number of months between today and '12-Jan-98'.
8. Find the experience of all the employees in terms of years.
9. Find the people who joined before '1987'.
10. Find the people who joined in April month.
11. Display the employee's names in sentence case, uppercase, lowercase.
12. Display first five characters of the string 'abcdfgh'.

13. Find the number of employees.
14. Find the number of departments in the organization.
15. Find the number of people who are working in department no 10.
16. Find the number of employees who are drawing the salary more than 3000.
17. Find the total salary, average salary, minimum, maximum salaries of all employee's salaries
18. Find the total salary of each employee (total sal=sal+comm. If comm is null consider min as 0).
19. Display employee name, salary and the message depending on the salary of employee
=1500 'on target' <1500 'less' >1500 'more'
20. Display the list of employees whose names contains 'a' irrespective of the case.
- 21. Use appropriate function and extract 3 characters starting from 2 characters from the following string 'Oracle' i.e. the output should be 'rac'.**
- 22. Find the first occurrence of character a from the following string 'computer maintenance corporation'.**
- 23. Replace every occurrence of alphabet A with B in the string Allen's (user translate function).**
- 24. Display the information from EMP table. Wherever job 'manager' is found it should be displayed as boss(replace function).**
- 25. Display those employees whose salary contains at least 4 digits?**

Staff In-charge

Mrs.R.Pallavi reddy
Mrs.K.Sindhura
Mrs.B.Spandana
Mrs.G.Amulya
Mr.P.Kiran

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LAB CYCLE – 2

Assignment – 1

Sub Queries and Correlated Nested Queries (in, any, all, exists, etc.)

1. Find the employees whose salary is more than the salary of 'SMITH'.
2. Find the no of employees whose salary is more than their manager's salary.
3. Find the employees whose salary is more than the average salary.
4. Find the list of 'SALESMAN' whose salary is equal to the maximum salary among them.
5. Find the employees whose experience is more than their managers.
6. Find the employees whose total salary (sal+comm) is more than the total salary (sal+comm.) of 'SMITH'.
7. Find the employees who earn minimum salary in each job and arrange them in descending order of salary.
8. Find the name of senior most employees in each department.
9. Display the departments that have no employees.
10. Write a query to display the message as 'early joined' and 'late joined' based on the hiredates of the employees comparing with '1-Jan-82'
11. Display the list of employees who earn more than every employee's salary in Department 30.
12. **Display the employee number and name of employee working as CLERK and earning highest salary among CLERKS.**
13. **Display the names of the salesman who earns a salary more than the highest salary of any clerk.**
14. **Display the names of employees who earn highest salaries in their respective job groups.**
15. **Display the employee names who are working in Chicago.**

Assignment – 2

Queries Using Aggregate operation, Group by and Having clauses

1. Find the number of employees in each department.
2. Display the lists of departments in which at least 4 employees are working.
3. Find the number of employees by department no, job.
4. Find the list of employees who are getting the maximum salaries in their respective departments.
5. Find the employees who are getting more than the average salary of their respective departments.
6. **Display the employee names who are working in accountings dept.**
7. **Display the employee names who are working in Chicago.**
8. **Display the job groups having total salary greater than the maximum salary for managers.**
9. **Display the names of employees from department number 10 with salary greater than that of any employee working in other departments.**
10. **Display the names of employee from department number 10 with salary greater than that of all employee working in other departments.**

Queries Using Joins expressions

11. Display employee names, department name in which they are working.
12. Display the list of employees who are working in the same city in which 'King' is working.
13. Display no of employees working in each city.
14. Display the city with the highest no of employees.
15. Display the no of employees departments wise in each city.
16. Display the list of employees who are working in either 'NEWYORK' or DELHI'
17. Find the list employees who are not staying in the same city in which their manager stays.
- 18. Display employee name, deptname, salary and comm. for those Sal in between 2000 to 5000 while location is Chicago.**
- 19. Display those manager who are not working under president but they are working under any other manager?**

Views

20. Create different types of views on tables.
21. Querying on created views.

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LAB CYCLE-3

Basic PL/SQL Programs, Functions and Procedures

1. Write a program to print the multiplication table of a given number up to 10 terms.
2. Write a program to print all prime numbers between a range.
3. Write a program to print the reverse number of a given number.
4. Write a program to update salary of the employee based on the following conditions
If salary is less than 2000 increment by 15%
>2000 & <3000 increase by 10% otherwise 10%.
5. Write a program that returns the total gross salary of a given department.
6. Write a program that returns no of employees, total salary of a given department.

Cursors

1. Write a program to print employee's details in the following format.
Emp Name total salary
Calculate total salary as (sal* 12+comm.*12).
2. Write a program to update the salary of the employee by 10% if the salary is less than 2000.

Triggers

1. Write a program for insert trigger
2. Write a program for delete trigger
3. Write a program for update trigger

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LAB CYCLE - 4

Case Study

A Case Study on any one real time database application

Example: Bus Reservation System:

Develop an online reservation system **Roadway Travels** database by using MySQL. This system should allow the users to do reservations by using credit or debit card through internet and can also enquire about the availability of buses, timings and seats.

The above process involves many steps.

1. Analyzing the problem and identifying the entities and relationships.
2. ER Model
3. Creation of Tables /relations as per the case study.
4. Normalization
5. Inserting data into the tables.
6. Practice the following Queries:
 - a. Display unique PPNO of all passengers.
 - b. Display all the names of male passengers.
 - c. Display the ticket numbers and names of all the passengers.
 - d. Find the names of passengers whose age is between 30 and 45.
 - e. Display the sorted list of passenger's names.
 - f. Write a Query to display the Information present in the Passenger and cancellation tables.
 - g. Find the total number of cancelled seats.
7. Design and develop the following:
 - i. Triggers
 - ii. Cursors

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